Patient Information Leaflet

hameln

Read all of this patient information leaflet carefully before you start taking this medicine.
- Keep this leaflet. You may need to read it again.
- If you have further questions, please ask your doctor or your pharmacist.
- This medicine has been prescribed for you personally and you should not pass it on to others. It may harm them, even if their symptoms are the same as yours.

Metoclopramide-hameln 5 mg/ml Injection

PRODUCT DESCRIPTION
Clear, colourless solution

COMPOSITION
Each 1 ml contains:
Metoclopramide hydrochloride 1 H_2O 5.268 mg
equivalent to metoclopramide HCl 5.000 mg
1 ampoule with 2 ml solution for injection contains
Metoclopramide hydrochloride 1 H_2O 10.536 mg
equivalent to metoclopramide HCl 10.000 mg

List of excipients:
Sodium chloride, citric acid monohydrate, sodium citrate dihydrate, water for injections, 1 N sodium hydroxide solution or 1 N hydrochloric acid (for pH adjustment)

1 ampoule of 2 ml solution for injection contains 6.37 mg sodium (= 0.28 mval sodium).

PROPERTIES
Pharmacodynamic properties:
Metoclopramide is a central dopamine antagonist. In addition it shows peripheral cholinergic activity. Two principal actions may be distinguished: (1) an antiemetic effect; (2) accelerated gastric emptying and small-intestinal transit.

The antiemetic effect is mediated by a central action on the brainstem (chemoreceptor trigger zone of the vomiting centre), probably via inhibition of dopaminergic neurones. The increased motility is also partly directed by higher centres, but a peripheral mechanism of action also operates via activation of postganglionic cholinergic receptors; inhibition of dopaminergic receptors in the stomach and small bowel may also play a part.

On prolonged administration, the lack of dopaminergic inhibition of prolactin secretion may lead to a rise in the serum prolactin concentration.

Pharmacokinetic properties:
Metoclopramide undergoes rapid initial distribution with onset of action between 1 to 3 minutes following intravenous administration and 10 to 15 minutes following intramuscular administration. Values between 2.6 and 4.6 hours have been determined for the elimination half-life. Metoclopramide shows little plasma protein binding. The distribution volume is between 2.2 and 3.4 l/kg.

Metoclopramide crosses the blood-brain barrier and is excreted in breast milk. It is excreted by the kidneys, partly unchanged (approx. 20%) and partly after metabolism in the liver where it undergoes conjugation to glucuronic or sulphuric acids.

INDICATIONS
Adult population:
Metoclopramide-hameln 5 mg/ml Injection, 2 ml is indicated for use in adults for:
- prevention of post-operative nausea and vomiting
- symptomatic treatment of nausea and vomiting, including nausea and vomiting induced by migraine attacks,
- prevention of chemotherapy- or radiotherapy-induced nausea and vomiting
- relief of nausea and vomiting associated with malignant disease, labour, infectious diseases and uraemia
- to assist in intestinal intubation

Pediatric population:
Metoclopramide-hameln 5 mg/ml Injection, 2 ml is indicated in children aged 1 to 18 years for:
- prevention of delayed chemotherapy-induced nausea and vomiting as a second-line option
- prevention of post-operative nausea and vomiting as a second-line option
- intolerance to cytotoxic drugs and, as an aid to intestinal intubation

Metoclopramide-hameln 5 mg/ml Injection is of little benefit for the prevention of treatment of motion sickness.

RECOMMENDED DOSAGE
The solution can be administered by the intravenous or intramuscular route.

The intravenous doses must be administered as a slow bolus (for at least 3 minutes).

Each 2 ml of Metoclopramide-hameln 5 mg/ml Injection contains 10 mg of Metoclopramide HCl anhydrous substance.

All indications (adults)
A single 10 mg dose is recommended for the prevention of post-operative nausea and vomiting.

The recommended dose for the symptomatic treatment of nausea and vomiting, including nausea and vomiting induced by migraine attacks and for the prevention of radiotherapy-induced nausea and vomiting is 10 mg per dose, 1 to 3 times daily. The maximum recommended daily dose is 30 mg or 0.5 mg/kg.

Treatment duration when administering by injection should be as short as possible and a switch to administration via oral or rectal route should be instituted as quickly as possible.

All indications (children from 1 to 18 years of age)
The recommended dosage is 0.1 to 0.15 mg/kg, 1 to 3 times daily, by intravenous route.

The maximum daily dose is 0.5 mg/kg.

Dosing Table

Age	Body weight	Dose	Frequency
1-3 years	10-14 kg	1 mg	Up to 3 times daily
3-5 years	15-19 kg	2 mg	Up to 3 times daily
5-9 years	20-29 kg	2.5 mg	Up to 3 times daily
9-18 years	30-60 kg	5 mg	Up to 3 times daily
15-18 years	Over 60 kg	10 mg	Up to 3 times daily

For the prevention of delayed chemotherapy-induced nausea and vomiting, the maximum treatment duration is 5 days.

For the prevention of post-operative nausea and vomiting, the maximum treatment duration is 48 hours.

Frequency of administration:
A minimum interval of 6 hours between two administrations is to be respected, even if vomiting or rejection of the dose occurs.

Special populations
Elderly subjects
In elderly subjects, a dose reduction should be considered, based on kidney and liver function, and overall frailty.

Kidney failure
In patients with end-stage kidney failure (creatinine clearance ≤ 15 ml/min), the daily dose should be reduced by 75%.

In patients with moderate to severe kidney failure (creatinine clearance between 15 and 60 ml/min), the dose should be reduced by 50%.

Liver failure
In patients with severe liver failure, the dose should be reduced by 50%.

Other pharmaceutical forms may be more suitable for these patient populations.

Pediatric population
Metoclopramide is contraindicated in children aged less than one year.

For diagnostic purposes:
A single slow IV dose may be given 10 minutes before the investigation.

Adults (20 years or older) and weighing 60 kg & above	10 – 20 mg
Young adults (15-19 years) and weighing 60 kg & above	10 mg
Young adults (15-19 years) with body weight of 30-59 kg	5 mg
Children (9-14 years) (30 to 59 kg)	5 mg
Children (5-9 years) 20-29 kg	2.5 mg
Children (3-5 years) 15-19 kg	2 mg
Children under 3 years Up to 14 kg	1 mg

Where body-weight is below that specified for a given age group, the dose should reflect the weight rather than the age, so that a lower dose is chosen.

Compatibility:
Intravenous fluids:
From the microbiological point of view, any dilutions should be used immediately. If the dilutions are not used immediately, in-use storage times and storage conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 2-8°C, unless dilution has taken place in controlled and validated conditions.

Metoclopramide-hameln may be added to the following solutions: Glucose 5% in sodium chloride 0.45%; glucose 5% in water; mannitol 20%; sodium chloride 0.9%; Ringer's Injection; Ringer's Injection, lactated (Hartmann's solution).

Any dilutions should be protected from light during infusion. Degradation is indicated by a yellow discoloration and such solution must not be used.

Cisplatin, cyclophosphamide and doxorubicin hydrochloride are stated to be compatible with metoclopramide hydrochloride but compatibility is dependent upon factors such as the particular formulation, drug concentration and temperature.

Incompatibility:
Proprietary preparations of metoclopramide hydrochloride are stated to be incompatible with cephalothin sodium, chloramphenicol sodium and sodium bicarbonate.

ROUTE OF ADMINISTRATION
The duration of administration of Metoclopramide-hameln 5 mg/ml Injection depends on the underlying disease. As a rule, a duration of 4-6 weeks is sufficient. Depending on the individual case, metoclopramide can be administered as required for up to 6 months.

Note: Prolonged treatment with Metoclopramide-hameln 5 mg/ml Injection is associated with an increased risk of the development of motoric disorders (see *Side effects*).

CONTRAINDICATIONS
Metoclopramide-hameln 5 mg/ml Injection must not be used in the following situations:
- Known hypersensitivity to the active ingredient metoclopramide or any of the other ingredients
- Gastrointestinal haemorrhage, mechanical obstruction or gastrointestinal perforation for which the stimulation of gastrointestinal motility constitutes a risk
- Confirmed or suspected pheochromocytoma, due to the risk of severe hypertension episodes
- History of neuroleptic or metoclopramide-induced tardive dyskinesia
- Epilepsy (increased crises frequency and intensity)
- Parkinson's disease
- Combination with levodopa or dopaminergic agonists
- Known history of methaemoglobinaemia with metoclopramide or of NADH cytochrome-b5 deficiency
- Use in children less than 1 year of age
- 3 to 4 days after gastrointestinal surgery
- Breast feeding

PRECAUTIONS AND WARNINGS
Neurological disorders
Extrapyramidal disorders may occur, particularly in children and young adults, and/or with doses higher than 0.5 mg per kg of body weight per day. Hence, doses exceeding 0.5 mg/kg/day should be avoided. These reac-

tions generally occur at the beginning of treatment, and can occur after a single dose. If extrapyramidal symptoms occur, metoclopramide should be discontinued immediately. These effects are generally completely reversible after treatment discontinuation; however, symptomatic treatment may be required (benzodiazepines in children, and/or anticholinergic antiparkinsonian medicinal products in adults).

An interval of at least six hours should be respected between each dose even if vomiting or rejection of the dose occurs, in order to avoid overdose.

Long-term treatment with metoclopramide may cause potentially irreversible tardive dyskinesia, particularly in elderly subjects. Treatment should not exceed 3 months because of the risk of tardive dyskinesia. Treatment must be discontinued if clinical signs of tardive dyskinesia occur.

Neuroleptic malignant syndrome has been described with metoclopramide in combination with neuroleptics and with metoclopramide monotherapy. Metoclopramide must be immediately discontinued if symptoms of neuroleptic malignant syndrome develop, and appropriate treatment should be initiated.

Particular caution should be exercised in patients with underlying neurological disorders, and in patients receiving other centrally-acting drugs.

Symptoms of Parkinson's disease may also be exacerbated by metoclopramide.

Methemoglobinemia

Methemoglobinemia, which could be related to NADH-cytochrome b5 reductase deficiency, has been reported. If this occurs, treatment must be immediately and permanently discontinued, and appropriate measures initiated (such as treatment with methylene blue).

Cardiac disorders

Serious cardiovascular undesirable effects, including cases of severe bradycardia, circulatory collapse, cardiac arrest and QT prolongation have been reported during administration of metoclopramide by injection, particularly via the intravenous route.

Particular caution should be exercised when administering metoclopramide, particularly via the intravenous route, in elderly subjects, patients with cardiac conduction disorders (including QT prolongation), patients with electrolyte imbalance, bradycardia, and patients taking other drugs known to prolong QT interval.

The intravenous injection must be given as a slow bolus (of at least 3 minutes duration) in order to reduce the risk of undesirable effects (e.g. hypotension, akathisia).

Kidney or liver failure

In patients with kidney failure or severe liver failure, a dose reduction is recommended.

Metoclopramide should be used with caution in patients with hepatic and renal impairment and in the elderly, young adults and children.

Because metoclopramide can stimulate gastroinstestinal mobility, the drug theoretically could produce increased pressure on the suture lines following gastrointestinal anastomosis or closure.

Metoclopramide should be used with caution in patients with hypertension, since there is limited evidence that the drug may increase circulating catecholamines in such patients.

Metoclopramide may mask underlying disorders such as cerebral irritation, epilepsy, pregnancy and porphyria. The frequency and severity of seizures may be increased by metoclopramide in patients with a history of seizure disorders.

If vomiting persists, the patients should be reassessed to exclude the possibility of an underlying disorders e.g. cerebral irritation.

DRUG INTERACTIONS

Metoclopramide-hameln 5 mg/ml Injection can alter the absorption of other substances; for example it can reduce the absorption of digoxin and cimetidine, and accelerate or increase the absorption of levodopa, paracetamol, various antibiotics (demonstrated in the case of tetracycline and pivampicillin), lithium and alcohol. When Metoclopramide-hameln 5 mg/ml Injection and lithium are administered concurrently, elevated plasma lithium levels may occur.

Anticholinergic agents can reduce the effect of Metoclopramide-hameln 5 mg/ml Injection.

When Metoclopramide-hameln 5 mg/ml Injection and neuroleptic agents (e.g. phenothiazines, thioxanthene derivatives, butyrophenones) are administered concurrently, aggravated extrapyramidal disorders (e.g. convulsive manifestations affecting the head, neck and shoulder region) may occur.

The concomitant administration of serotonin-reabsorption hemmers can also lead to amplified occurrences reaching from extrapyramidal symptoms to serotonin syndrome.

The effect of succinylcholine may be prolonged by Metoclopramide-hameln 5 mg/ml Injection.

Pregnancy and lactation

As there are no adequate and controlled studies using metoclopramide in pregnant women, metoclopramide should be used during pregnancy only when clearly needed and is not recommended in the first trimester.

Since metoclopramide is distributed into milk, the drug should be used with caution in nursing mothers.

Effects on ability to drive and use machines

Even when used appropriately, this drug may alter reaction capacity to such an extent as to impair the capacity to drive in traffic or operate machines. This applies particularly when used in combination with alcohol or sedative agents.

SIDE EFFECTS

Dystonic reactions include oculogyric crisis, spasm of the facial or masticatory muscles, protrusion of the tongue, trismus, unnatural positioning of the head and the neck, overstretching of the spinal column, shortening reactions of the arms and extension spasm of the legs. Should treatment of a dystonic reaction is required, an anticholinergic, anti-parkinsonian drug or a benzodiazepine may be used.

Long-term administration may lead to raised prolactin levels and this may lead to gynaecomastia, galactorrhoea or menstrual disturbances.

Cases of malignant neuroleptic syndrome have been occurring very rarely during administration of metoclopramide. Characteristic signs are: fever, muscular rigidity, alterations in consciousness and blood pressure. The following immediate measures are recommended: discontinuation of treatment with Metoclopramide-hameln 5 mg/ml solution for injection, cooling, dantrolene and/or bromocriptine, adequate fluid administration.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Symptoms of overdosage include somnolence, confusion, irritability, restlessness, convulsions, extrapyramidal motor disturbances, disturbances of cardiovascular function with bradycardia and rises or falls in blood pressure. There have been a few reports of the occurrence of methaemoglobinaemia in premature and full term neonates who were given overdoses of metoclopramide (1 to 4 mg/kg/day orally, intramuscularly or intravenously for 1 to 3 or more days).

Treatment of metoclopramide overdosage generally involves symptomatic and supportive care. There is no specific antidote for metoclopramide, however, agents with central anticholinergic activity (e.g. diphenhydramine, benstropine) may be useful in extrapyramidal reactions. The patient should be treated with gastric lavage. Symptoms of metoclopramide overdose are generally self-limiting and usually subside within 24 hours. Haemodialysis or peritoneal dialysis is unlikely to enhance the elimination of metoclopramide.

PACKING OR PACK SIZES

Box of 5 or 10 ampoules with 2 ml solution for injection.

STORAGE CONDITIONS

Do not store above 30°C. Protect from light.

Once removed from its packaging, the drug should not be subjected to prolonged exposure to direct sunlight.

Any unused solution from opened ampoules should be discarded.

Keep out of reach of children.
Jauhi daripada kanak-kanak.

Shelf life

Please refer to the outer box label.

Manufacturer

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